

Notes: Ioannidou, Pavanini & Peng (JFE, 2022)

Collateral and Asymmetric Information in Lending Markets

Contents

1	Big picture	3
1.1	Incremental contribution of the paper	3
2	Data construction and institutional setting	4
2.1	Bolivian credit registry and sample window	4
2.2	Loan-level variables	4
2.3	Collateral values and recovery assumptions	5
2.4	Bank-level and market variables	5
3	Structural model and empirical specifications	5
3.1	Borrower demand for secured and unsecured loans	5
3.2	Default model	6
3.3	Joint distribution of unobservables and adverse selection	6
3.4	Bank profit and equilibrium pricing	6
3.5	Prediction of unobserved contract menus and prices	6
4	Identification strategy	7
4.1	Step 1: construct the choice set	7
4.2	Step 2: identify demand using variation in contract terms	7
4.3	Step 3: identify adverse selection from demand-default covariance	7
4.4	Step 4: identify moral hazard from default response to terms	7
4.5	Step 5: identify bank supply and marginal costs	8
4.6	Threats to identification and responses	8
5	Tables and figures	8
5.1	Table 1: Summary Statistics of Commercial Loans and Figure 1: Number of Banks Establishing Lending Relationships with New Borrowers across Regions	8
5.2	Figure 2: Propensity Score of Choosing a Secured Loan and Figure 3: Collateral to Loan Ratio by Types	9
5.3	Table 2: Price Prediction for Average Interest Rate and Table 3: Loan and Borrower Characteristics for Different Contract Types	10
5.4	Figure 4: Price Prediction Accuracy and Table 4: First Stage Results	11
5.5	Table 5: Structural Estimation Results and Figure 5: Random Coefficients of Price and Collateral	11
5.6	Table 6: Descriptives of Model Fit and Figure 6: Model-Implied Marginal Costs and Banks' Funding Costs	12
5.7	Table 7: Determinants of Marginal Cost and Profit Margin and Table 8: Secured Lending, Adverse Selection, and Moral Hazard	13

5.8	Table 9: The Collateral Channel and Table 10: The Determinants of Credit Rationing	14
5.9	Table 11: Effectiveness of Collateral as a Screening Device	14
6	Interpretive synthesis	15
7	Short summary	15
8	Conclusion	16
8.1	Core conclusion	16
8.2	Economic intuition	16
8.3	Incremental contribution for the literature	16
8.4	What the paper does not fully settle	16
8.5	Final takeaway	16

1 Big picture

- **Question.** What are the benefits and costs of collateral in bank lending markets when borrowers have private information about credit risk and banks can use both prices and collateral requirements to screen borrowers?
- **Main setting.** The paper studies commercial loans in Bolivia using the public credit registry between 1999 and 2003. The setting is useful because information asymmetries are severe, collateral is sometimes used, and the registry reports loan terms and ex post performance.
- **Core mechanism.** Collateral can help banks screen borrowers ex ante because safer borrowers dislike collateral less and interest rates more. Collateral can also reduce ex post moral hazard because pledged assets make default less attractive. But collateral can also amplify shocks: when collateral values fall, banks may raise rates, ration borrowers, and reduce credit supply.
- **Model contribution.** The paper builds a structural model of borrower demand for secured and unsecured loans, loan amount choice, borrower default, bank contract availability, and bank pricing. The model allows collateral to affect both borrower selection and repayment incentives.
- **Identification idea.** The model combines demand choices, ex post default outcomes, predicted contract menus and prices, and instruments for interest rates and collateral requirements. Correlations among unobserved demand and default components identify adverse selection; direct effects of rate and collateral on default identify moral hazard.
- **Main evidence.** Riskier borrowers are less price-sensitive and more collateral-sensitive. Safer borrowers are more willing to post collateral in exchange for lower interest rates. Collateral decreases default probability, consistent with ex post discipline.
- **Counterfactual result.** Removing secured lending reduces credit demand and bank profits. A collateral value shock raises secured-loan rates, reduces loan sizes and demand, increases default, and lowers bank profits when banks respond only by pricing.
- **Rationing result.** If banks can respond through both pricing and rationing, they reject borrowers most damaged by the collateral shock and preserve collateral's screening role for borrowers least affected by the shock.
- **Economic takeaway.** Collateral is not simply a debt-capacity device. It is a screening and incentive instrument, but its usefulness depends on collateral values and on whether banks can adjust contract menus, not only interest rates.

1.1 Incremental contribution of the paper

- **Unifies benefits and costs of collateral.** Prior empirical work often isolates either credit demand or credit supply. This paper places both in a single equilibrium framework, so the same model can evaluate collateral's screening role, its incentive role, its price effects, its rationing effects, and its effects on bank profitability.
- **Separates adverse selection from moral hazard.** The paper uses the joint distribution of unobserved preferences for interest rates and collateral, together with ex post default, to identify adverse selection. It then uses direct rate and collateral effects in the default equation to capture moral hazard after selection is controlled for.
- **Models secured and unsecured credit jointly.** Existing structural work on asymmetric information in lending often studies unsecured credit. This paper extends the framework to a differentiated loan market where the same bank can offer secured loans, unsecured loans, both, or neither.
- **Adds contract availability and rationing.** The paper does not assume every borrower receives every possible contract. Banks can screen by not offering some contracts. This is crucial in the collateral shock counterfactual, where rationing preserves screening for remaining borrowers.

- **Quantifies spillovers from secured to unsecured credit.** A shock to collateral values directly hits secured lending, but it also changes unsecured loan pricing and demand because borrowers substitute across contracts and banks reoptimize prices.
- **Policy relevance.** The analysis shows that policies aimed at collateral values, recovery rates, credit registries, or bank screening cannot be evaluated from one margin alone. Demand, pricing, default incentives, and rationing interact.

2 Data construction and institutional setting

2.1 Bolivian credit registry and sample window

- The data come from the Central de Informacion de Riesgos Crediticios (CIRC), the public credit registry supervised by the Bolivian Superintendent of Banks and Financial Entities.
- The registry covers all formal licensed financial institutions and reports borrower-bank loan contracts, which is valuable in a market where audited statements and private credit bureau information were limited.
- The usable sample starts in March 1999, because loan type is observed from that date onward. Earlier information from January 1998 is used to identify pre-existing bank-borrower relationships and validate price prediction.
- The main sample follows firms taking loans for the first time during the sample period and tracks them over the first 18 months. This reduces informational asymmetry between the econometrician and banks from long-standing relationships.
- The focus is on commercial term loans from commercial banks. This keeps loan types and lender types relatively homogeneous and focuses on loans where collateral is sometimes, but not always, pledged.

Interpretation: The key empirical design feature is that the paper studies borrowers at the start of formal lending relationships. These borrowers are most likely to face information frictions, so collateral should matter most. At the same time, the credit registry provides ex post repayment outcomes needed to connect choices to default risk.

2.2 Loan-level variables

- For each loan the registry reports the originating bank, origination date, maturity date, loan amount, interest rate, loan type, collateral type and value, and repayment performance.
- Loan types include secured and unsecured commercial term loans. Collateral is classified into immovable collateral, liquid movable collateral, and illiquid movable collateral.
- The paper measures the value-to-loan ratio for secured loans and constructs collateral indicators. The modal secured loan has collateral value close to the loan amount, especially when movable collateral is used.
- Default and non-performing loan indicators come from ex post repayment information.
- Relationship lending is measured with a dummy indicating whether the borrower already had a prior lending relationship with the bank.

Interpretation: The loan-level variables are enough to estimate borrower choice and ex post default, but they do not reveal all contracts offered to each borrower. That missing choice set is why the paper needs a contract availability and price prediction procedure.

2.3 Collateral values and recovery assumptions

- Collateral is not modeled as a free choice by borrowers. The paper assumes borrowers present pledgeable assets and banks determine the type and amount of collateral required.
- For secured loans, recovery depends on collateral value and on the collateral type. The model uses expected recovery rates of 100% for immovable assets, 90% for liquid movable assets, and 70% for illiquid movable assets.
- For unsecured loans, the recovery rate is inferred from observed loss rates among defaulting borrowers in the same industry and credit rating group.
- The paper documents that borrowers appear collateral constrained. Borrowers with secured first loans are less likely to obtain additional secured loans shortly afterward, and additional secured loans typically use the same collateral type.

Interpretation: These assumptions matter for the collateral shock counterfactual. If borrowers could easily pledge other assets after a value drop, the shock would be muted. The evidence suggests that many firms are already using their pledgeable assets intensively.

2.4 Bank-level and market variables

- The registry and balance sheet data provide bank-level deposit information, including saving deposit interest rates.
- The paper uses the saving deposit interest rate as an instrument for loan interest rates, interpreting it as a proxy for bank funding costs.
- The share of non-performing loans in a bank's existing portfolio, by geographic market and previous quarters, is used as an instrument for collateral requirements.
- Regional and quarter variation matter because banks differ across markets in funding, risk exposure, and contract availability.

Interpretation: The bank and market variables are used to address endogeneity in prices and collateral. They help separate variation in contract terms driven by bank costs or portfolio risk from variation driven by borrower unobservables.

3 Structural model and empirical specifications

3.1 Borrower demand for secured and unsecured loans

A borrower i considering bank j , loan type $k \in \{S, U\}$, and market m receives indirect utility:

$$U_{ijk}^D = \alpha_P^D P_{ijk} + \alpha_C^D C_{ijk} + \alpha_Z^D Z_{ijm} + X_{jm} \alpha_X^D + \xi_{ijk} + \varepsilon_{ik}^D,$$

where P_{ijk} is the interest rate, C_{ijk} indicates collateral, Z_{ijm} captures an existing bank-borrower relationship, and X_{jm} includes observed bank-market controls.

Interpretation: Demand responds to both price and collateral. The random components in price and collateral sensitivity capture unobserved borrower heterogeneity, which is essential for identifying asymmetric information.

3.2 Default model

Borrower default under a specific loan is modeled with indirect utility:

$$U_{ijk}^F = \alpha^F + \alpha_{Pi}^F P_{ijk} + \alpha_{Ci}^F C_{ijk} + \alpha_Z^F Z_{ijm} + X_{jm} \alpha_X^F + Y_i \alpha_Y^F + \varepsilon_i^F.$$

The price and collateral effects can depend on the relationship dummy:

$$\alpha_{Pi}^F = \alpha_P^F + \alpha_{PZ}^F Z_{ijm}, \quad \alpha_{Ci}^F = \alpha_C^F + \alpha_{CZ}^F Z_{ijm}.$$

Interpretation: The direct effects of interest rate and collateral on default capture ex post moral hazard. A positive price effect on default means higher debt burden worsens repayment incentives. A negative collateral effect means pledged assets discipline borrowers.

3.3 Joint distribution of unobservables and adverse selection

The demand and default unobservables follow a multivariate normal distribution:

$$\begin{pmatrix} \varepsilon_{Pi}^D \\ \varepsilon_{Ci}^D \\ \varepsilon_i^F \end{pmatrix} \sim N \left(0, \begin{pmatrix} \sigma_P^2 & \rho_{PC} \sigma_P \sigma_C & \rho_{PF} \sigma_P \\ \rho_{PC} \sigma_P \sigma_C & \sigma_C^2 & \rho_{CF} \sigma_C \\ \rho_{PF} \sigma_P & \rho_{CF} \sigma_C & 1 \end{pmatrix} \right).$$

Interpretation: A positive ρ_{PF} means riskier borrowers are less sensitive to price, so they are more likely to take credit. A negative ρ_{CF} means riskier borrowers dislike collateral more, so collateral screens them out. A negative ρ_{PC} means price-sensitive borrowers tend to be collateral-tolerant, creating a tradeoff banks can use to design separating contracts.

3.4 Bank profit and equilibrium pricing

Expected bank profit from loan type k is:

$$\Pi_{ijk} = [(1 + T_{ijm} P_{ijk})(1 - F_{ijk}) - MC_{ijk} + R_{ijk} F_{ijk}] Q_{ijk},$$

where F_{ijk} is default probability, MC_{ijk} is marginal cost, R_{ijk} is recovery in default, and Q_{ijk} is expected loan demand.

For secured loans:

$$R_{ijSm} = \min\{CV_{ijm} \omega_{ijSm}, 1 + T_{ijm} P_{ijSm}\}.$$

For unsecured loans:

$$R_{ijUm} = \omega_{ijUm} (1 + T_{ijm} P_{ijUm}).$$

Interpretation: Banks choose contract menus and interest rates taking into account borrower demand, default risk, recovery, and marginal costs. This makes the model an equilibrium model of credit supply rather than a reduced-form demand equation.

3.5 Prediction of unobserved contract menus and prices

The paper observes chosen loans but not all offers. It predicts potential offers in three steps.

- First, it estimates average loan interest rates with loan controls, bank-market fixed effects, and borrower fixed effects.
- Second, it uses propensity score matching to infer counterfactual secured and unsecured rates for similar borrowers.
- Third, it combines average rate predictions and propensity-score differences to predict secured and unsecured rates for each borrower-bank pair.

Interpretation: This price prediction procedure is central because borrower demand can only be estimated if the econometrician knows the alternatives borrowers faced. The paper validates this approach by showing that predicted prices are close to observed prices and do not predict default once borrower fixed effects are used.

4 Identification strategy

4.1 Step 1: construct the choice set

The econometrician does not observe all loan offers, so the first identification task is to infer which banks and contracts are available. A bank is included in a borrower's choice set if it is active in the relevant region-quarter market and grants similar loans. Banks may offer secured loans, unsecured loans, both, or neither.

Interpretation: The key assumption is that availability can be proxied from observed bank activity among comparable borrowers. This allows the model to distinguish not choosing a contract from not being offered that contract.

4.2 Step 2: identify demand using variation in contract terms

Borrower demand is identified from how borrowers choose among banks and contract types with different interest rates, collateral requirements, relationships, amounts, and maturities.

Interpretation: Demand elasticities are not purely reduced-form correlations because prices and collateral may be endogenous. The paper uses control functions based on instruments to purge variation in interest rates and collateral requirements that is correlated with borrower unobservables.

4.3 Step 3: identify adverse selection from demand-default covariance

Adverse selection is identified from the relationship between unobserved borrower preferences and ex post default risk. If borrowers with high unobserved default risk are less price-sensitive, the estimate of ρ_{PF} should be positive. If risky borrowers dislike collateral more, ρ_{CF} should be negative.

Interpretation: This is the same conceptual logic as positive correlation tests for asymmetric information in insurance markets, but adapted to a differentiated credit market with collateral and pricing. The model uses borrowers' choices and defaults jointly, not just default rates on observed contracts.

4.4 Step 4: identify moral hazard from default response to terms

After selection on unobservables is modeled, the direct effect of interest rates and collateral in the default equation identifies ex post incentive effects. Higher rates should raise default, while collateral should lower default if it disciplines borrowers.

Interpretation: This distinction is important: a high observed default rate on unsecured loans could come from selection of risky borrowers or from weaker repayment incentives. The structural model separates the two channels.

4.5 Step 5: identify bank supply and marginal costs

The bank pricing first-order conditions connect observed or predicted prices to default probabilities, demand probabilities, recovery, and marginal costs. The model backs out marginal costs consistent with Bertrand-Nash pricing.

Interpretation: The supply side is identified by imposing banks' optimal pricing conditions. This allows counterfactuals to include not only borrower reactions but also bank price and rationing responses.

4.6 Threats to identification and responses

- **Unobserved offers.** The paper does not observe rejected offers. It addresses this with contract availability prediction and out-of-sample price validation.
- **Price endogeneity.** Banks may price based on private borrower risk. The paper uses saving deposit rates as instruments and adds control-function residuals.
- **Collateral endogeneity.** Banks may require collateral for risky borrowers. The paper instruments collateral with prior non-performing loan shares.
- **Relationship lending.** Existing relationships may reveal soft information to banks. The paper includes relationship terms and focuses on new borrowers to reduce information asymmetry between banks and econometrician.
- **Collateral constraint assumption.** The counterfactual assumes firms cannot pledge additional assets after a collateral shock. The paper supports this with evidence that secured borrowers rarely add new secured loans with different collateral shortly afterward.

5 Tables and figures

5.1 Table 1: Summary Statistics of Commercial Loans and Figure 1: Number of Banks Establishing Lending Relationships with New Borrowers across Regions

Read:

- Table 1 compares first loans of new borrowers with a broader active-borrower sample. Interest rates are around 14%, secured loans have lower average rates than unsecured loans, and new borrowers use collateral more often than the broader sample.
- Panel B reports bank-level deposit variables; Panel C reports loss rates. Unsecured loans to defaulting borrowers have meaningful loss rates, which inform unsecured recovery assumptions.
- Figure 1 maps the number of banks establishing lending relationships with new borrowers across Bolivian regions. Bank availability is geographically uneven.

Interpretation: Table 1 establishes the basic facts needed for the model: secured and unsecured loans coexist, collateral is not universal, and default/recovery outcomes can be measured. Figure 1 shows why local bank availability and regional fixed effects matter for choice sets.

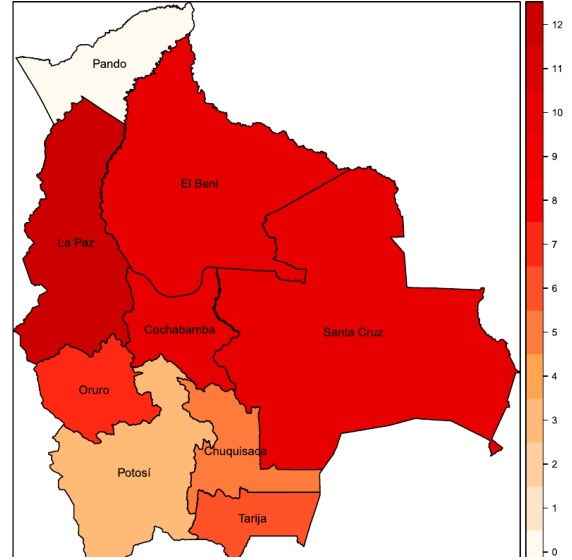
Economic message: Collateral is used in a setting with heterogeneous bank competition and recovery prospects, making it possible to estimate how contract terms screen borrowers.

Table 1
Summary Statistics of Commercial Loans.

Variable	N. Obs	Mean	St. Dev.	N. Obs	Mean	St. Dev.
Panel A: Loan Level						
New Borrowers			Borrowers Active since 6 Months			
Interest Rate	1650	14.29	2.62	9400	14.33	2.32
Secured	519	13.80	2.40	2185	13.66	2.53
Unsecured	1131	14.51	2.68	7215	14.53	2.22
Collateralized	1650	0.31	0.46	9400	0.23	0.42
Immovable	519	0.41	0.49	2185	0.39	0.49
Liquid Movable	519	0.30	0.46	2185	0.26	0.44
Illiquid Movable	519	0.29	0.46	2185	0.35	0.48
Value-to-Loan Ratio	519	2.66	4.42	2185	2.49	6.04
Amount	1650	146.58	401.12	9400	129.84	428.76
Maturity	1650	15.49	21.95	9400	12.63	18.05
Installment Loan	1650	0.55	0.50	9400	0.50	0.50
Bad Credit Rating	1650	0.04	0.19	9400	0.12	0.32
Corporation	1650	0.65	0.48	9400	0.65	0.48
Defaulting Borrower	1650	0.12	0.32	9400	0.26	0.44
Panel B: Bank Level						
Saving Deposit	619	62.17	51.78			
Demand Deposit	619	60.10	46.05			
Deposits to Liabilities	619	0.73	0.12			
Saving Deposit Interest Rate	619	6.99	3.32			
Panel C: Loss Rates						
Loss Given Default	283	0.35	0.46			
Secured	134	0.29	0.45			
Unsecured	149	0.41	0.48			
Loss from Defaulting Borrower	299	0.05	0.18			

Note: This table summarizes information from three datasets we use. Panel A's unit of observation is a new borrower's first loan or a loan granted to borrowers who entered the credit registry since no more than 6 months. Interest Rate is the annual percentage rate, which is divided into two subgroups: interest rate for secured loans (Secured) and unsecured loans (Unsecured). Collateral is a dummy variable taking the value of one if a loan is secured and zero if it is unsecured. Immovable is a dummy variable taking the value of one if the collateral is immovable (real estate) and zero otherwise. Movable (Illiquid (Movable Liquid)) is a dummy variable taking the value of one if the collateral is movable but (illiquid (movable and liquid)), for example, inventory, equipment, vehicle, accounts receivable (for example, bank deposits, bonds, securities), and zero otherwise. Value-to-Loan Ratio is the ratio of collateral value to the loan amount for secured loans only. The loan Maturity is in months, and loan Amount is in 1000 USD. Installment is a dummy variable taking the value of one if this is an installment loan and zero if it is a single payment loan. Bad Credit Rating is a dummy variable taking the value of one if the loan has any overdue payments or is in default and zero otherwise. Corporation is a dummy variable taking the value of one if the borrower is a corporation and zero if it is a sole proprietorship or partnership. Defaulting Borrower is a dummy variable taking the value of one for loans that are granted to borrowers who had at least one non-performing loan within the sample period and zero otherwise. Panel B's unit of observation is a bank-month level balance sheet entry. Saving Deposit, Demand Deposit are in millions of USD. Saving Deposit Interest Rate is the annual percentage rate. Panel C's unit of observation is a defaulted loan or a loan granted to a defaulting borrower. Loss Given Default is the loss rate of defaulted loans. Loss from Defaulting Borrower is the loss rate of defaulted loans or a loan granted to a defaulting borrower.

(a) Table 1: Summary statistics of commercial loans



(b) Figure 1: bank relationships by region

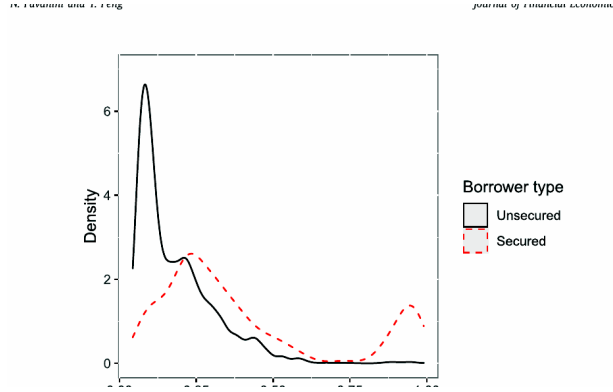
5.2 Figure 2: Propensity Score of Choosing a Secured Loan and Figure 3: Collateral to Loan Ratio by Types

Read:

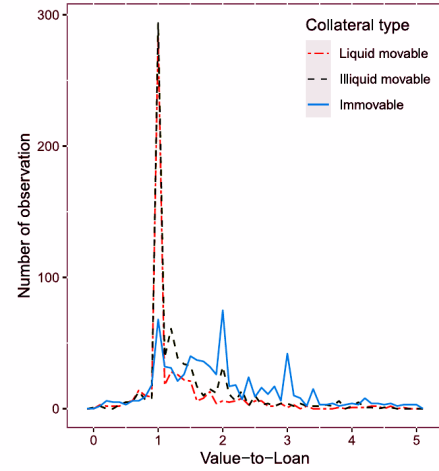
- Figure 2 shows that the propensity scores of secured and unsecured borrowers overlap substantially. This supports matching borrowers who choose different contract types.
- Figure 3 shows collateral-to-loan ratios by collateral type. Movable collateral is concentrated around value-to-loan equal to one, while immovable collateral often has higher value-to-loan ratios.

Interpretation: Figure 2 is important for the price-prediction procedure: it shows common support between secured and unsecured borrowers. Figure 3 motivates using collateral type and recovery assumptions rather than treating all collateral as identical.

Economic message: The market contains comparable borrowers choosing different contract types and heterogeneous collateral quality, creating scope for screening.



(a) Figure 2: propensity score of choosing secured loan



(b) Figure 3: collateral-to-loan ratio by type

5.3 Table 2: Price Prediction for Average Interest Rate and Table 3: Loan and Borrower Characteristics for Different Contract Types

Read:

- Table 2 shows that borrower fixed effects greatly improve price prediction; the adjusted R^2 is high for observed prices with borrower fixed effects.
- Price residuals do not predict default once borrower fixed effects are included, suggesting that predicted prices are not missing borrower risk in a way that would bias the default model.
- Table 3 shows that borrowers offered both contracts have lower default incidence than those offered only one contract type, and that only-secured borrowers often demand larger or longer loans.

Interpretation: Table 2 validates the construction of unobserved prices, while Table 3 validates the idea that banks use contract availability to screen borrowers. Together they support the demand estimation and choice-set assumptions.

Economic message: The model requires credible counterfactual prices and menus. These tables show that the predicted environment looks economically plausible.

Table 2
Price Prediction for Average Interest Rate.

	Borrower FE		No Borrower FE	
	Observed Price	Default	Observed Price	Default
Price Residual		-0.002 (0.01)		0.04*** (0.003)
Loan Controls	Yes	Yes	Yes	Yes
Borrower Controls	No	Yes	Yes	Yes
Bank-Market FE	Yes	Yes	Yes	Yes
Borrower FE	Yes	No	No	No
Observations	9400	9400	9400	9400
Adjusted R^2	0.85	0.19	0.63	0.21

Note: This table shows the price prediction for average interest rate. The first column shows the OLS regression result for Eq. (12). The dependent variable is observed interest rate. Loan controls include fixed effects for loan amount and maturity categories, and a dummy for installment loan. Borrower controls include dummies for bad credit rating, corporation, and industry. The second column is to show the price prediction does not miss determinants for default. The price residual means the residuals from Eq. (12). The dependent variable is the indicator for Non-performing. The third and fourth column repeat the exercise with no borrower fixed effects in the model * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. More details on this regression are reported in Table A.4 of the Appendix.

(a) Table 2: price prediction for average interest rate

Table 3
Loan and Borrower Characteristics for Different Contract Types.

	Both	Only Unsecured	Only Secured
Loan Amount	128,221	122,494	165,775
Maturity	11.30	10.49	14.67
Corporation	0.67	0.66	0.65
Bad Credit Rating	0.13	0.13	0.14
Interest Rates	13.47	13.87	13.91
Manufacturing	0.30	0.28	0.24
Construction	0.09	0.08	0.08
Wholesale and Retail Trade	0.28	0.26	0.29
Real Estate Activities	0.04	0.04	0.03
Social Services	0.01	0.02	0.02
Other Activities	0.27	0.33	0.33
Observations	39,778	28,379	7976

Note: This table summarizes average characteristics of borrowers and of loans that were offered by all available banks. A bank may offer both secured and unsecured contracts to a borrower (Both), only an unsecured loan (Only Unsecured), or only a secured loan (Only Secured). Amount is the loan amount in USD. Maturity is in months. Corporation is a dummy variable taking the value of one if the borrower is a corporation and zero if it is a sole proprietorship or partnership. Bad Credit Rating is a dummy variable taking the value of one if the loan has any overdue payments or is in default and zero otherwise. Interest Rate is the annual percentage rate. Manufacturing, Construction, Wholesale and Retail Trade, Real Estate Activities, Social Services, Other Activities are dummy variables indicating borrowers' industry.

(b) Table 3: loan and borrower characteristics by contract type

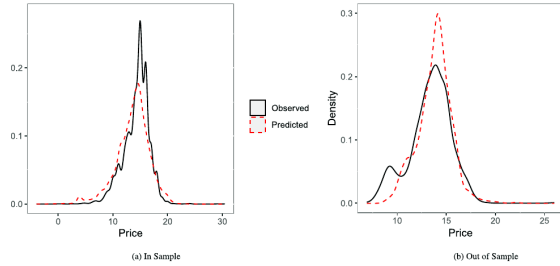
5.4 Figure 4: Price Prediction Accuracy and Table 4: First Stage Results

Read:

- Figure 4 compares observed and predicted interest-rate distributions in sample and out of sample. Predicted rates track observed rates reasonably well.
- Table 4 reports the first stage for interest rates and collateral. Saving deposit interest rates predict loan rates, and the share of non-performing loans predicts collateral requirements.

Interpretation: Figure 4 shows that the price-prediction exercise is not only internally fitted but also has out-of-sample content. Table 4 supports instrument relevance, which is needed for the control-function strategy.

Economic message: The identification of demand and default relies on predicted prices and instruments; these results are the empirical backbone of the model.



4. Price Prediction Accuracy. Note: This figure shows the distributions of observed prices (solid line) and predicted prices (dashed line). Subfigure (a) shows the in-sample prediction. The total number of observations is 9,400. Subfigure (b) shows the out-of-sample prediction. The total number of observations is 103,137.

(a) Figure 4: price prediction accuracy

Table 4
First Stage Results.

	Price		Collateral	
	Observed	Predicted	Observed	Predicted
Savings Deposit Interest Rate	0.30*** (0.01)	0.10*** (0.01)		
Share of Non-Performing Loans			1.04*** (0.08)	0.11*** (0.03)
Loan Controls	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Relationship FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes
Observations	8938	103,137	9190	105,883
Adjusted R ²	0.45	0.10	0.25	0.06

Note: This table shows the first stage results for prices and collateral using the first 18 months of each borrower in the original sample. In the first column, the dependent variable is the price we observe in the sample. In the second column, the dependent variable is the predicted price. Similarly, in the third column, the dependent variable is the collateral dummy observed in the sample, while in the fourth column, the dependent variable is the collateral dummy resulting from our prediction model. The instrumental variable is the interest rate of household saving deposits for interest rates, and the share of non-performing loans in banks' outstanding loans for collateral. Loan Controls include dummy variables for installment, Corporation, Bad Credit Rating, amount and maturity categories. The number of observations for prices is less than the total number of predicted and observed contracts due to some missing values of the instrumental variables. There

(b) Table 4: first-stage results

5.5 Table 5: Structural Estimation Results and Figure 5: Random Coefficients of Price and Collateral

Read:

- Table 5 shows that demand falls with interest rates and collateral requirements. The loan amount also falls with rates and reacts to collateral.
- In the default equation, higher interest rates increase default probability and collateral decreases default probability.
- The covariance matrix implies positive correlation between price sensitivity and unobserved default risk, negative correlation between collateral sensitivity and unobserved default risk, and negative correlation between price and collateral sensitivities.
- Figure 5 visualizes this selection pattern: riskier borrowers cluster where collateral disutility is high and price disutility is low.

Interpretation: This is the core evidence of the paper. The positive price-default selection correlation indicates adverse selection into borrowing; the negative collateral-default correlation indicates that collateral screens risky borrowers out of secured loans; the direct default effects indicate moral hazard and repayment incentives.

Economic message: Collateral works because it is costly for risky borrowers and valuable to lenders in default. It simultaneously affects who borrows and how borrowers behave after borrowing.

Table 5
Structural Estimation Results.

	Demand	Loan Size	Default
Price	-0.43*** (0.01)	-0.22*** (0.01)	0.93*** (0.01)
Price × Relationship	-1.15*** (0.00)	0.03 (0.03)	0.22*** (0.01)
Collateral	-0.50*** (0.01)	0.44*** (0.12)	-0.11*** (0.01)
Collateral × Relationship	-0.10*** (0.01)	0.28* (0.14)	0.44*** (0.02)
Installment		-0.06 (0.10)	-0.20*** (0.01)
Corporation		0.30*** (0.07)	0.04*** (0.01)
Bad Credit Rating		0.78*** (0.18)	0.96*** (0.04)
Bank FE	Yes	Yes	Yes
Relationship FE	Yes	Yes	Yes
Amount FE	No	No	Yes
Maturity FE	No	Yes	Yes
Industry FE	No	Yes	Yes
Region FE	No	Yes	Yes
Price residual	Yes	Yes	Yes
Collateral residual	Yes	Yes	Yes
Observations	16,652 $\sigma_D = 1.75^{***}$ (0.01)	1650	1650
Covariance matrix	$\rho_{PC} = -0.60^{***}$ (0.00) $\rho_{PD} = 0.10^{***}$ (0.01)	$\sigma_C = 0.36^{***}$ (0.01) $\rho_{CY} = -0.42^{***}$ (0.01)	$\sigma_Y = 1$

Note: This table presents the estimates of the structural model. The first column is for demand and the third column is for default, all of which are estimated using maximum simulated likelihood. The second column is for loan amount estimated using a two-step regression with the control function approach, where the dependent variable of the second step is the logarithm of loan amount. There are two random coefficients in demand, price and collateral, with mean coefficient reported in the first column of results and with standard deviation coefficients reported in the covariance matrix panel. In the demand

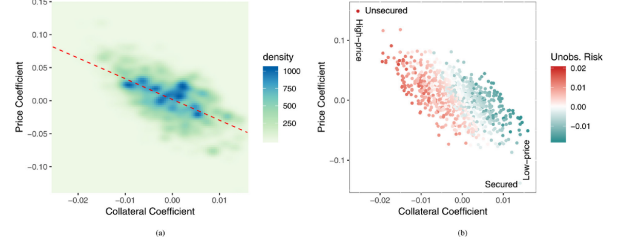


Fig. 5. Random Coefficients of Price and Collateral. Note: These figures plot model estimated heterogeneity in price and collateral coefficients for all loans. Subfigure (a) plots the joint density of the random price and collateral coefficients. The dashed line is the linear model fitted line which captures the correlation between the heterogeneous price and collateral coefficients. Subfigure (b) plots each observation explicitly. *Unobs. Risk* is the estimated unobserved risk. High unobserved risk firms are in red and low unobserved risk firms are in green. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(a) Table 5: structural estimation results

(b) Figure 5: random coefficients of price and collateral

5.6 Table 6: Descriptives of Model Fit and Figure 6: Model-Implied Marginal Costs and Banks' Funding Costs

Read:

- Table 6 compares actual data with model-implied baseline equilibrium outcomes. Predicted interest rates, default, demand, and profits are close to observed values.
- The model implies marginal costs around 8 percentage points and profit margins around 5 percentage points.
- Figure 6 compares model-implied marginal costs with bank funding costs over time. Both series decline over the sample, supporting the reasonableness of the supply-side estimates.

Interpretation: Model fit is not only about matching choices; it must also produce reasonable implied costs. Table 6 and Figure 6 help validate the supply side that will be used for counterfactuals.

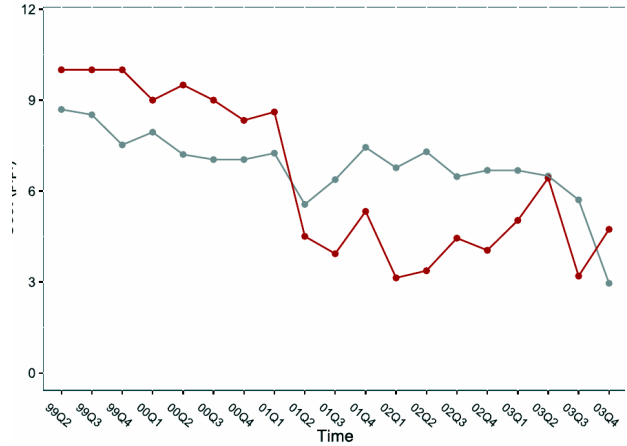
Economic message: The model can replicate observed market outcomes and generate plausible marginal costs, making the policy counterfactuals more credible.

Table 6
Descriptives of Model Fit.

	N. Obs	Mean	Median	Std. Dev
Actual Interest Rate	14,338	13.37	13.76	3.37
Baseline Interest Rate	14,313	13.12	13.55	3.68
Actual Default	14,338	0.09	0.08	0.09
Baseline Default	14,313	0.09	0.08	0.09
Actual Demand	14,338	7644	0.00	24,242
Baseline Demand	14,313	8720	0.00	27,133
Actual Profit	14,338	507	0.00	2878
Baseline Profit	14,313	524	0.00	3048
Marginal Cost	14,313	8.02	8.33	3.52
Profit Margin	14,313	5.34	5.16	1.40

Note: This table summarizes the model fit results. For each variable we report both descriptive statistics from the data (Actual) and the model predicted equilibrium in the baseline scenario (Baseline). Interest Rate is in percentage points. Default is a probability. Demand is the product of demand probability and loan amount in USD. Marginal Cost is the annualized cost of lending in percentage points. Profit Margin is the spread between the interest rate and the marginal cost.

(a) Table 6: descriptives of model fit



(b) Figure 6: model-implied marginal costs and funding costs

5.7 Table 7: Determinants of Marginal Cost and Profit Margin and Table 8: Secured Lending, Adverse Selection, and Moral Hazard

Read:

- Table 7 shows secured loans have higher marginal costs and lower profit margins. Offering both secured and unsecured contracts is also costly, consistent with screening and monitoring costs.
- Table 8 evaluates counterfactuals shutting down secured lending, adverse selection, or moral hazard.
- Banning secured lending eliminates secured contracts, shifts borrowers to unsecured loans, raises unsecured demand, but reduces total demand and bank profits.
- Eliminating moral hazard sharply reduces default probabilities and raises profits, indicating moral hazard is quantitatively important.

Interpretation: Table 7 shows that collateralized lending is not free for banks. Table 8 shows why collateral matters: it expands access and bank profitability, and it is closely tied to both adverse selection and moral hazard.

Economic message: Collateral is beneficial despite higher bank costs because it improves sorting and repayment incentives.

	Marginal Cost		Profit Margin	
	(1)	(2)	(3)	(4)
Collateral	4.45*** (1.53)	3.82*** (1.58)	-4.98*** (1.54)	-4.33*** (1.58)
Both		2.88 (1.76)		-2.96* (1.76)
Loan Controls	Yes	Yes	Yes	Yes
Borrower Controls	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes
Observations	16,852	16,852	16,852	16,852
Adjusted R ²	0.03	0.03	0.03	0.03

Note: This table shows the OLS regression results of model implied marginal costs and of the profit margins on different loan and firm characteristics. The dependent variable in Columns (1) to (2) are the normalized marginal costs, and in Columns (3) to (4) are the profit margin, i.e., the difference between interest rates and marginal costs. Collateral is an indicator which equals one for a secured loan and zero for an unsecured one. Both equals one if a loan belongs to a pair of secured and unsecured loans that are offered by one bank to the same borrower. Loan Controls include variables for Amount, Maturity, Installment, Rat Credit Ratios, Relationships. Borrower Controls include variables for Concentration, Industry. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

(a) Table 7: marginal cost and profit margin

		Median Percentage Change				
		Available Contracts	Interest Rate	Default	Loan Size	Demand
(A) No Secured Loans						
Secured	0	NA	NA	NA	-100	-100
Unsecured	9575	-1.26	-1.57	1.24	62.74	52.71
Total	9575	-0.00	-0.00	-25.50	-16.65	-17.96
(B) No Adverse Selection						
Secured	7277	0.41	1.17	-0.40	-1.65	-0.73
Unsecured	9575	-0.41	-0.44	0.23	1.25	0.53
Total	16,852	-0.04	0.07	-0.11	-0.11	-0.06
(C) No Moral Hazard						
Secured	7277	0.29	-66.68	-0.45	-2.47	0.00
Unsecured	9575	-1.42	-78.21	1.45	3.49	35.39
Total	16,852	-0.41	-75.27	0.00	-0.00	17.65

Note: This table summarizes the median percentage change in equilibrium price (Interest Rate), probability of default (Default), loan size (Loan Size) expected demand (Demand) and banks' expected profit (Profit) of loans in three counterfactual scenarios compared with the baseline model. Each row stands for secured loans, unsecured loans, and both. Panel A, B, and C present the counterfactual without secured lending, adverse selection, and moral hazard, respectively.

(b) Table 8: secured lending, adverse selection, and moral hazard

5.8 Table 9: The Collateral Channel and Table 10: The Determinants of Credit Rationing

Read:

- Table 9 simulates a 40% drop in collateral values. If banks only reprice, secured loan rates rise, default rises, loan size and demand fall, and bank profits fall.
- If banks can also ration borrowers, many contracts are withdrawn; changes in rates, defaults, demand, and profits are much smaller for remaining borrowers.
- Table 10 shows that loans with higher expected recovery rates are less likely to be rationed and loans to borrowers with bad credit ratings are more likely to be rationed.

Interpretation: Table 9 is the collateral-channel counterfactual. Its key message is that pricing alone propagates collateral shocks through higher rates and lower demand, while rationing reallocates credit away from borrowers for whom collateral no longer screens effectively. Table 10 explains which borrowers and loans are rationed.

Economic message: Rationing can be privately rational and stabilizing for banks, but it also restricts credit access for borrowers most damaged by collateral-value declines.

Table 9
The Collateral Channel.

	Available Contracts	Median Percentage Change				
		Interest Rate	Default	Loan Size	Demand	Profit
(A) Pricing						
Secured	7277	2.62	3.03	-12.28	-2.13	-6.51
Unsecured	9575	0.00	0.00	-4.30	0.00	-1.06
Total	16,852	1.37	2.75	-10.43	-3.78	-6.63
(B) Pricing & Rationing						
Secured	4742	0.00	0.00	-0.28	-0.00	-1.59
Unsecured	5549	0.00	-0.21	0.00	0.00	0.00
Total	10,291	0.00	-0.04	0.00	-0.00	-0.00

Note: This table summarizes the median percentage change in equilibrium price (Interest Rate), probability of default (Default), loan size (Loan Size) expected demand (Demand) and banks' expected profit (Profit) of loans in two counterfactual scenarios compared with the baseline model. Each row stands for secured loans, unsecured loans, and both. The two panels present the scenarios after a 40% collateral value drop compared with the baseline model. Panel A presents the case where the set of available contracts is fixed, whereas Panel B shows the rationing case, where loans with negative expected profits are not offered to borrowers. The percentage changes in interest rate (probability of default, or demand) are calculated by comparing the average interest rate (probability of default, or demand) across loan offers in each borrower's choice set, weighted by their demand probabilities in the counterfactual and in the baseline scenario. The percentage changes in loan size is calculated by comparing the average loan size across offers in each borrower's choice set. The

(a) Table 9: the collateral channel

Table 10
The Determinants of Credit Rationing.

	Credit Rationing
Expected Recovery Rate	-0.08* (0.05)
Bad Credit Rating	0.21*** (0.06)
Borrower Controls	Yes
Bank FE	Yes
Region FE	Yes
Quarter FE	Yes
Observations	16,852

Note: This table shows how the characteristics of loans and borrowers determine the likelihood of credit being rationed after the collateral value shock. The column reports marginal effects of a probit model, where the dependent variable takes value of one if a loan has negative profit after the collateral value shock, and zero otherwise. *Expected Recovery Rate* is the expected recovery rate in default, defined in Eqs. (9) and (10). *Bad Credit Rating* is a dummy variable taking the value of one if the borrower has a loan with any overdue payment or default. *Borrower Controls* include variables for Corporation, Industry. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

more likely to be rationed when collateral value drops. We

(b) Table 10: determinants of credit rationing

5.9 Table 11: Effectiveness of Collateral as a Screening Device

Read:

- Table 11 regresses the probability of choosing a secured loan on simulated unobserved borrower risk.
- In the baseline, unobserved risk is negatively related to choosing a secured loan. This means safer borrowers are more likely to select into secured loans.
- Under a collateral shock with pricing only, the screening coefficient weakens and loses significance.
- With pricing and rationing, the negative screening coefficient reappears strongly among remaining borrowers.

Interpretation: The table demonstrates the interaction between collateral values and contract menus. Collateral screens only when collateral values are high enough and when banks can withdraw contracts for borrowers whose collateral no longer supports effective separation.

Economic message: The screening role of collateral is endogenous to market conditions. A collateral shock can destroy screening unless banks can ration.

Table 11
Effectiveness of Collateral as a Screening Device.

	Probability of Choosing Secured Loan		
	Baseline	Collateral Shock with Screening via	
		Pricing	Pricing & Rationing
Unobserved Risk	-0.55** (0.24)	-0.41 (0.35)	-0.37*** (0.12)
Borrower Controls	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes
Region FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Observations	5784	5784	3364
Adjusted R ²	0.03	0.05	0.10

Note: This table summarizes OLS regression results. The unit of observation is a borrower-bank combination, conditional on the bank offering both secured and unsecured loans to the borrower. Interest Rate is the interest rate of the secured loan. The dependent variable is the conditional probability of choosing the secured contract from the pair of contracts provided by the same bank. The explanatory variable *Unobserved Risk* is the simulated unobserved risk. The three columns correspond to baseline model, collateral channel with pricing response, and collateral channel with pricing and rationing response. *Borrower Controls* include variables for Corporation, Industry. We add a dummy variable to control for loan contracts with negative expected profit, and exclude contracts with extremely low demand probabilities (the median demand probability of these excluded contracts is 8.47e-11.) * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 11: effectiveness of collateral as a screening device

6 Interpretive synthesis

The paper’s central synthesis is that collateral is a multi-purpose contract term. It increases expected recovery in default, separates safer from riskier borrowers, and disciplines repayment incentives. These effects are jointly determined. A secured loan is not just a loan with higher recovery; it is a different product that attracts different borrowers and changes repayment incentives.

The counterfactuals show that collateral-value shocks operate through multiple margins. When collateral values fall, banks’ expected recovery falls, secured loans become less attractive to banks, interest rates rise, borrowers substitute between secured and unsecured contracts, default risk changes, and some borrowers are rationed. The aggregate effect depends on the bank’s full response, not just the direct fall in collateral value.

The most policy-relevant result is the distinction between pricing and rationing. Pricing alone passes collateral shocks into interest rates and can weaken screening. Pricing plus rationing rejects the worst affected borrowers and preserves the screening value of collateral among remaining borrowers. This is privately useful for banks, but it also means collateral shocks reduce access to credit for some firms.

7 Short summary

- The paper studies collateral in lending markets with asymmetric information using Bolivian credit registry data.
- It estimates a structural model of secured and unsecured loan demand, loan size, default, bank contract menus, and pricing.
- The data focus on new commercial borrowers between 1999 and 2003, when information frictions were severe.
- Borrowers with high unobserved risk are less price-sensitive and more collateral-sensitive.
- Safer borrowers are more likely to choose secured loans when they receive lower rates in exchange for collateral.
- Higher interest rates increase default; collateral decreases default.

- These patterns support both adverse selection and moral hazard theories of collateral.
- Secured lending is costly for banks but expands credit access and improves profitability relative to banning secured loans.
- A 40% collateral value shock raises secured loan rates and reduces demand and profits when banks only reprice.
- When banks can ration, many contracts are removed, and the remaining contracts preserve collateral's screening role.
- The incremental contribution is to quantify both benefits and costs of collateral in one equilibrium framework that includes demand, supply, default, and contract availability.

8 Conclusion

8.1 Core conclusion

Ioannidou, Pavanini, and Peng show that collateral has large effects on credit access, pricing, default, and bank profitability. It mitigates adverse selection because risky borrowers dislike pledging collateral more than safe borrowers. It mitigates moral hazard because collateral reduces default incentives. But its effectiveness depends on collateral values and on banks' ability to adjust contract menus.

8.2 Economic intuition

The economic intuition is a separation mechanism. Safe borrowers are willing to post collateral in exchange for lower interest rates; risky borrowers prefer unsecured loans with higher rates because they are more likely to default and dislike losing pledged assets. Banks can exploit this preference difference by offering secured and unsecured contracts. When collateral values fall, this separation weakens because the collateral no longer provides enough recovery or discipline. Rationing then becomes a way to preserve separation for borrowers whose collateral remains valuable.

8.3 Incremental contribution for the literature

The paper goes beyond reduced-form collateral-shock studies by estimating all major moving parts of the lending market. It also goes beyond prior structural asymmetric-information models by introducing secured and unsecured loans, bank screening through both price and collateral, endogenous contract availability, and bank rationing. This combination lets the authors quantify not only whether collateral matters, but why it matters and how its role changes in crisis-like collateral shocks.

8.4 What the paper does not fully settle

The paper does not directly observe all rejected offers, all pledgeable assets, or all bank private information. It models loan size and maturity in a restricted way and holds some bank funding-cost conditions fixed in counterfactuals. Therefore, the estimates should be read as model-based quantification of key mechanisms rather than a purely quasi-experimental causal estimate.

8.5 Final takeaway

Collateral is a powerful but state-dependent screening and incentive device. In normal times it expands lending and improves sorting. In collateral downturns it can amplify credit contractions unless banks can ration contracts and preserve the screening role for borrowers whose collateral remains strong.